Please read the instructions below before the exam starts, and listen carefully to the instructions given by the invigilators.

You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2024

EC115 2023/4 A 800
EC115 2022/3 A 800
EC115 2022/3 A 850

Formula Sheet
Statistical Tables

Only Casio FX-83GTPLUS or FX-85GTPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted

UNIVERSITY OF READING

Introductory Quantitative Methods in Economics and Business (EC115)

Two Hours

Answer All questions.

- Clearly indicate the question number while writing an answer on your answer script.
- Show all the steps required to reach the solution. Even if your answer is correct, marks will be deducted if you only write the final answer without showing any calculations.
- Calculators are allowed. But do not use calculators for probability calculations. Instead, use the distribution tables provided in the Formula sheet for calculation of probabilities.

1. Expected value

A casino is contemplating two games that have the following prizes and probabilities. The tables show win and loss amounts in GBP (indicated with X) for the players of the games and the respective probabilities (Indicated with P(X=x) corresponding to each x).

Game A	
X (£)	P(X=x)
-15.00	0.4
25.00	0.1
32.00	0.1
10.00	0.1
-10.00	0.3

Game B	
X (£)	P(X=x)
-8.00	0.3
5.00	0.2
-4.00	0.2
10.00	0.1
1.00	0.2

(a) Assuming both games get played by the same number of players over the year, which game is expected to bring in higher profit to the casino owner?

(5 marks)

(b) During an informal survey by the Casino owner, an inexperienced player argues that they will always play Game A because the average amount a player can win from Game A is £8.4, while from Game B it is only £0.8. Is the player making a correct argument? Explain your answer.

(4 marks)

[Hint: You have to explain <u>why</u> the argument is right or wrong.]

(c) Which game has a higher variance?

(7 marks)

(d) Suppose the owner is also thinking of introducing a new rule by which a player has to pay a £1 fee for each game they wish to play but all their earnings (or losses) will also be inflated by 10% (i.e., a gain of £25 will have a bonus of £2.5 but a loss of £15 will also require paying another £1.5 from the player's pocket). Which of the two games will be more profitable from a player's perspective? Which game will have lower variability in outcomes?

(4 marks)

2. Probability and normal distribution

Suppose X is a normally distributed random variable with a mean of 125 and a variance of 576. Obtain the following probabilities and show the steps involved in the calculation.

(a) $P(X < 136)$ (2 marks)

(b) $P(X > 90)$ (2 marks)

(c) $P(111 < X < 122)$ (3 marks)

(d) Find b such that $P(X < b) = 0.67$ (3 marks)

3. Aggregate data

The table below provides summary data on the distribution of the salaries of employees in non-managerial roles within a company.

Department	Number of employees	Average gross salary (£/Year)
R&D	12	34,000
Production	30	27,500
Distribution	15	25,800
Marketing	10	32,600
Accounting	8	36,000
HR	5	28,500

(a) Calculate the average gross annual salary of an employee in this company.

(3 marks)

(b) Compute the standard deviation (rounded up to 2 decimal points is good) and indicate its unit of measurement.

(5 marks)

(c) A rival company shows a coefficient of variation of 20%. Which company shows a higher relative dispersion in the distribution of income - the given company or the rival?

(2 marks)

4. Conditional probability

Consider the following contingency table with probabilities concerning an accused in a murder case being tried at a certain district court. The letters in the parentheses are used to indicate the respective events. Use these same notations in all your answers.

	Convicted (C)	Not Convicted (NC)
Guilty (G)	0.40	0.10
Not guilty (NG)	0.15	0.35

(a) Using the notations in the above table, formally express the numbers in the table as probabilities. (e.g., the number 0.40 in the above table represents a probability – probability of what? Write as P(?,?) where the question marks should be replaced by the appropriate letters denoting different events.)

(2 marks)

(b) Find the marginal probability of being guilty and the marginal probability of being convicted.

(2 marks)

(c) Given the accused is actually guilty, what is the probability that the judicial process will be able to convict him?

(4 marks)

(d) Suppose the court acquits the accused, i.e., the accused isn't convicted. What is the probability that he is indeed innocent?

(4 marks)

(e) Consider the joint event of the accused being guilty as well as convicted. What is the complement of this event? Answer in plain language as well as using formal notation. Find the probability of this complement event.

(4 marks)

(f) Suppose a forensic test has 80% chance of correctly identifying the perpetrator. A positive test implies that the accused is the perpetrator. Therefore, if T indicates the event that the test is positive, then P(T|G)=0.80. There is also a 3% chance of the test being positive when the accused is innocent, i.e., P(T|NG) = 0.03. If the probability of the accused being guilty or not remains the same as before, then calculate the probability of the test being positive.

(4 marks)

(g) Find the probability of the accused being guilty given a positive forensic test.

(4 marks)

(h) Fill in the probabilities in the following contingency table. Remember that the probability of the accused being guilty or not is still the same as before.

	Forensic test positive (T)	Forensic test negative (NT)	Marginal probabilities
Guilty (G)			
Not guilty (NG)			
Marginal probabilities			

You may either produce a similar table on your answer script or write the probabilities after identifying each entry with formal notation as you did in question 4.(a).

(4 marks)

(i) Do you think the forensic test can improve the quality of judgement? Justify your answer assuming that conviction is 100% based on the test result.

(2 marks)

5. Inference

Consider this sample of 10 cauliflowers produced by a farm.

Table: Sample of cauliflowers from a farm

Cauliflower No.	Weight (grams)
1	588
2	610
3	553
4	640
5	612
6	580
7	610
8	530
9	647
10	630

The farm claims to produce cauliflowers with an average weight of 615 grams. A supermarket is using this sample to consider if they should buy the cauliflowers from the farm at a piece rate or priced per kilogram.

(a) Write down the null and alternative hypotheses the company should use to evaluate whether the farm's claim is correct and prescribe a statistical test to verify the claim at 1% significance level. What do they conclude? Show the workings and interpret your findings.

(10 marks)

(b) Obtain the same answer by computing the appropriate confidence interval. Show the workings. How confident are you that the average weight of a cauliflower from this farm lies in this interval?

(4 marks)

(c) How would the confidence interval change if you assume a 5% significance level? Interpret the difference between the confidence intervals obtained with 1% and 5% significance levels, respectively.

(6 marks)

(d) If the same average and variance used in (a), (b) and (c) above were obtained from a very large sample of 1000 cauliflowers, how would the confidence interval you computed in (c) for 5% level of significance change? Will the company's conclusion about the farm's claim remain the same as before?

(5 marks)

(e) Will the company have sufficient reason to reject the farm's claim at 5% level of significance if it is evaluating the same given sample of 10 cauliflowers, but believes the population standard deviation to be 28 grams?

(5 marks)

(End of question paper)